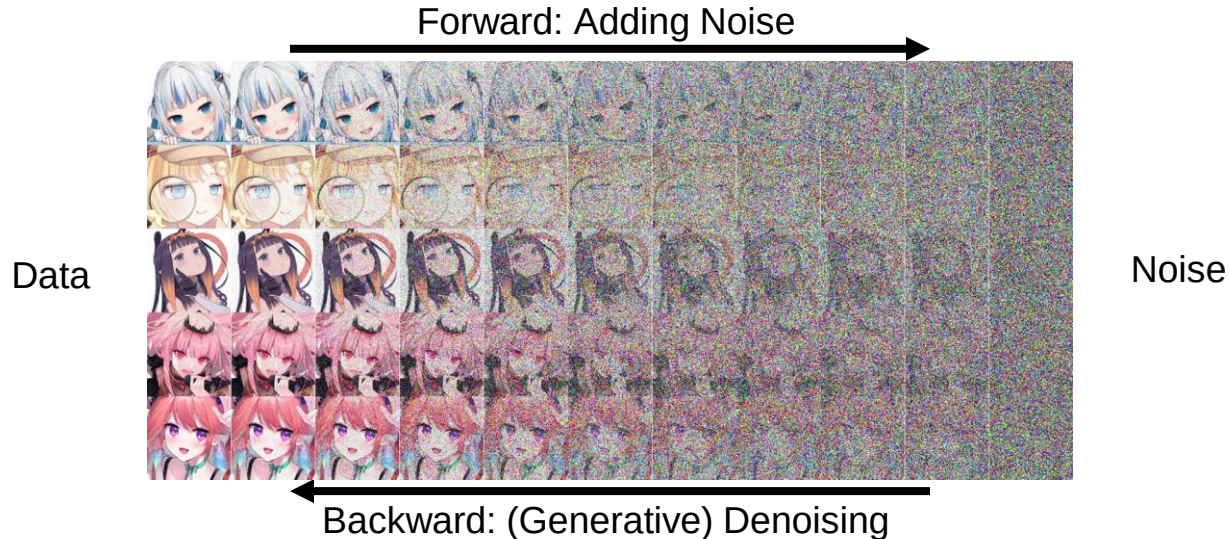


Homework 5 – Generative AI

Due Date: 2025/06/13 17:00 (UTC+8)

Introduction

In this assignment, the focus is on diffusion models in the field of generative AI, which generate images from a batch of noise. The process consists of two stages: the forward process and the backward process.

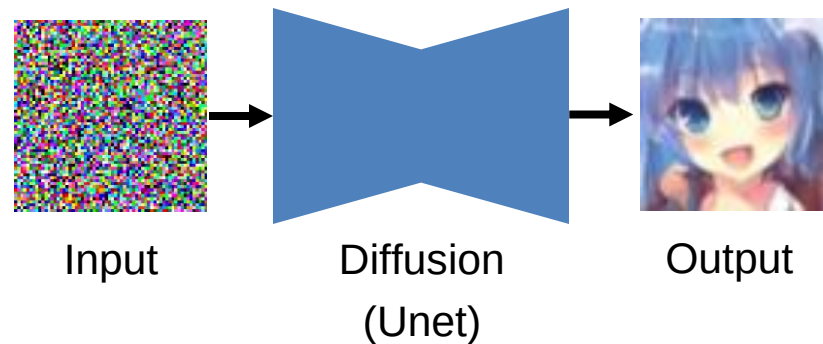


Introduction

Part 1 – Anime Face Generation

In this part, you will learn how to train a diffusion model from scratch. The given dataset contains 71,314 anime face images.

- Given: dataset
- Todo: diffusion model
- Target: 1000 anime face images

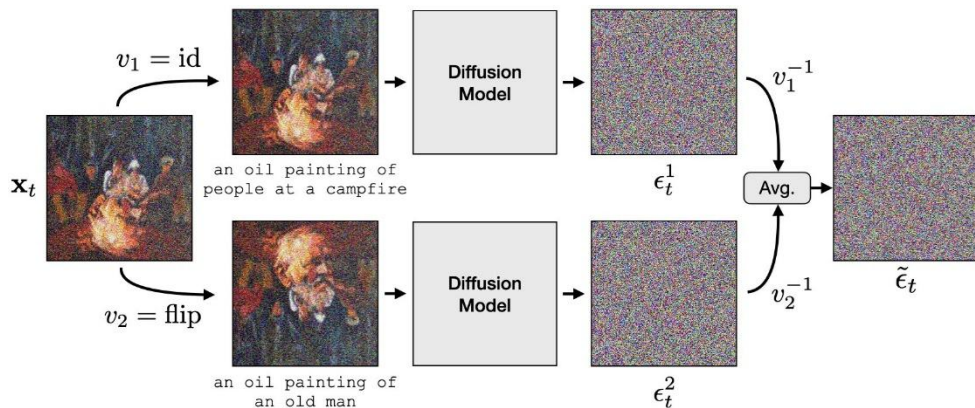


Introduction

Part 2 – Optical Illusion Generation

In this part, you will learn how to use existed text-to-image diffusion model. In addition, you need to change the denoising operation to realize the effect of optical illusion.

- Given: pretrained diffusion model
- Todo: denoising operation
- Target: an optical illusion image



Requirements

1. Please modify the codes in [source code](#) between `# Begin your code` and `# End your code`.
2. This assignment requires **high-level** hardware resource (GPU), it's recommended to run your code on [Colab](#) or [Kaggle](#).
3. It's **forbidden** to use additional data, pretrained weights, or other packages in your implementation.

For more detail requirements, please check the spec!

Implementation (70%)

- Part 1-1: Denoising Process (15%)
 - Implement **initialization, forward and backward function** in **diffusion.ipynb**
- Part 1-2: Result Visualization (10%)
 - Visualize the **denoising progress** and plot the **loss curve** in **diffusion.ipynb**
- Part 1-3: Evaluation Baseline (20%)
 - In this section, your generated images must meet a certain standard, including **FID Score** and **AFD rate**.

Implementation (70%)

Standard	FID Score	AFD Rate	Grade
Default	≤ 160.0	≥ 0.60	0%
Simple	≤ 120.0	≥ 0.70	5%
Normal	≤ 100.0	≥ 0.80	5%
Medium	≤ 90.0	≥ 0.85	5%
Hard	≤ 80.0	≥ 0.90	3%
Boss	≤ 70.0	≥ 0.95	2%

Implementation (70%)

- Part 2-1: Prompt Design (5%)
 - Please design **at least 2 different prompts** for different views in **multi-view.ipynb**
- Part 2-2: Viewing Transformation (5%)
 - Implement **IdentityView()** and **Rotate180View()** in **multi-view.ipynb**
- Part 2-3: Denoising Operation (10%)
 - Apply viewing transformation on **denoising process** in **multi-view.ipynb**
- Part 2-4: Evaluation Baseline (5%)
 - In this section, your generated images must meet a certain **CLIP Score (>0.3)** before and after image transformation. More details please refer to our spec.

Report (30%)

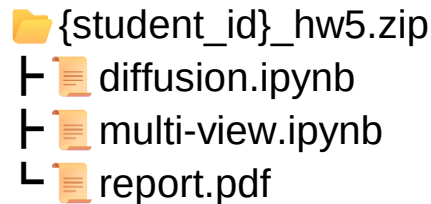
- You should write your report using the provided report template.
- The report should be written in **English**.
- Please save the report as a **.pdf** file. (font size: 12)
- For each part of the assignment, please take some screenshots of your code and provide a detailed explanation of your implementation.

Submission

Due Date: 2025/06/13 17:00

Please compress your [source code](#), and [report \(.pdf\)](#) into

{STUDENTID}_hw5.zip. The file structure should look like:



```
graph TD; A["{student_id}_hw5.zip"] --> B["diffusion.ipynb"]; A --> C["multi-view.ipynb"]; A --> D["report.pdf"]
```

The diagram shows a folder icon representing the zip file {student_id}_hw5.zip. Inside this folder, there are three file icons: diffusion.ipynb, multi-view.ipynb, and report.pdf, each preceded by a file icon.

Wrong submission format leads to -10 point.

Late submissions will not be accepted.

QA Page

If you have any questions about this homework, please ask them on the E3 Forum page. We will answer them as soon as possible. Additionally, we encourage you to answer other students' questions if you have any idea 😊

Reminder

When you begin to work on this assignment, it's recommended to use **Ctrl+F+“Todo”** to find each part for implementation.

Please note that training a diffusion model is **highly time-consuming!** Each training round may take **over an hour**.

It's best to start this assignment as soon as possible.
Good Luck!

Me: Waiting for the diffusion model training to complete



Reference

[1] Machine Learning Material from Hung-Yi Lee

<https://reurl.cc/RLp2xg>

[2] Denoising Diffusion Probabilistic Models (NeurIPS 2020)

<https://arxiv.org/pdf/2006.11239>

[3] Denoising diffusion implicit models (ICLR 2021)

<https://arxiv.org/pdf/2010.02502>

[4] DeepFloyd IF on Hugging Face

https://huggingface.co/docs/diffusers/api/pipelines/deepfloyd_if

[5] Visual Anagrams (CVPR 2024 oral)

<https://arxiv.org/pdf/2311.17919>

Please check out the spec
for more details!

Appendix – Kaggle Usage

How to modify code on Kaggle

- (1) Click the code link from spec
- (2) Click “Copy & Edit” for Kaggle
- (3) Choose “GPU T4 x2”
- (4) Start session and run the code
- (5) Download Notebook and Rename!!!

Introduction

Generative AI (Gen AI) has become a trending topic worldwide, especially after the release of ChatGPT. Gen AI can generate various types of content, including text, images, audio, and video. Recently, in the field of image generation, a novel technique called the Diffusion Model has gained significant attention, which this assignment focuses on. In this task, you will implement the diffusion process and train your own diffusion model from scratch. Additionally, you will generate a special-effect image using a pre-trained text-to-image diffusion model. This special effect requires an additional denoising operation without requiring further training or fine-tuning.

The goal of this programming assignment is to:

- 1) Understand the architecture and design of U-Net.
- 2) Implement a diffusion model and train it from scratch.
- 3) Learn how to evaluate generative models (e.g., FID score).
- 4) Utilize existing (diffusion) models available on Hugging Face.
- 5) Explore a training-free method for diffusion.

Requirements

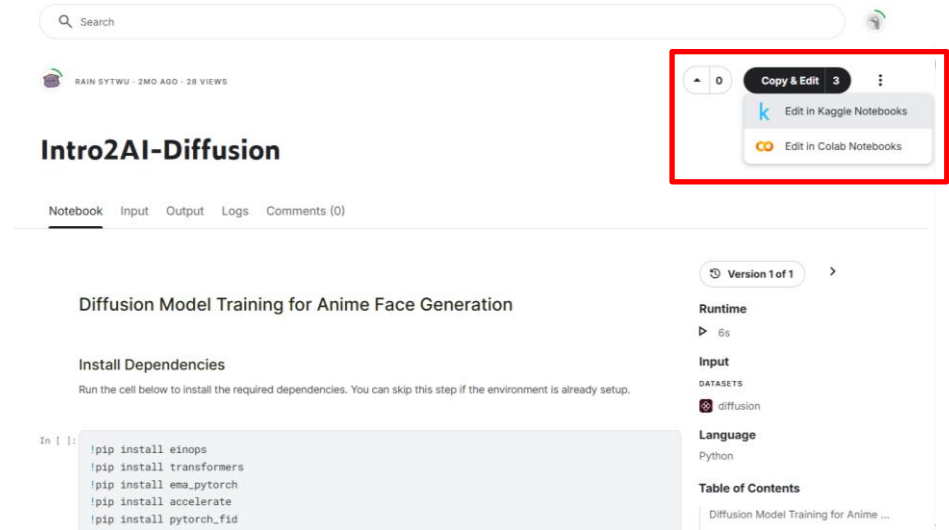
1. Please modify the code in [source code](#) between [# Begin your code](#) and [# End your code](#). If you modify the other code (e.g., [Add HorizontalFlip\(\)](#) for data augmentation), please specify in the corresponding section or appendix.
2. This assignment requires [high-level](#) hardware resources (GPU), we suggest running your code on [Colab](#) or [Kaggle](#). We recommend using [Kaggle](#) since it allows a free account to access [T4 GPU up to 30 hours](#) weekly.

Codes on Kaggle Notebook: [Part1](#) | [Part2](#) | [FID&AFD Evaluation](#)

3. [Do not use](#) extra data, pre-trained weights, or external packages. Submitting the dataset as output may result in [zero points](#). Ensure your code is reproducible; significant differences in our results will [reduce your score](#).

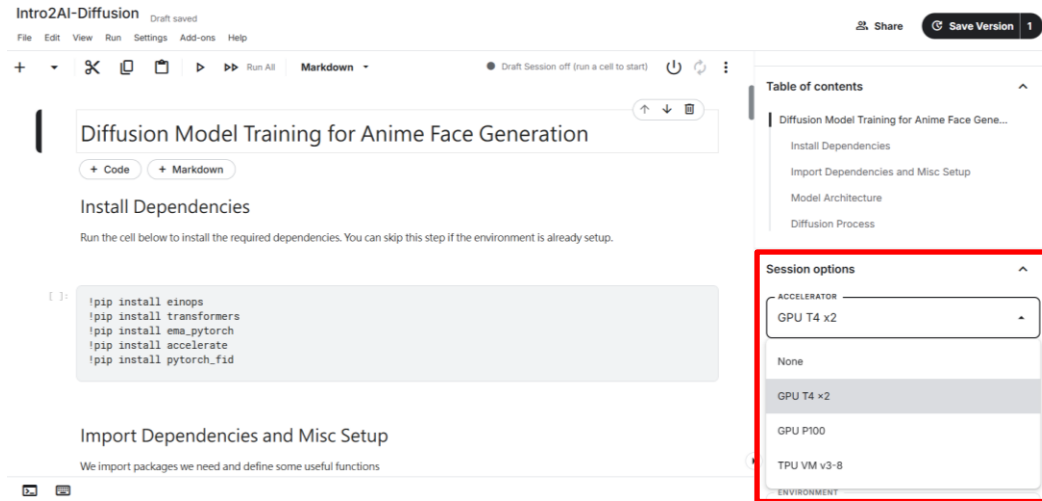
How to modify code on Kaggle

- (1) Click the code link from spec
- (2) Click “Copy & Edit” for Kaggle
- (3) Choose “GPU T4 x2”
- (4) Start session and run the code
- (5) Download Notebook and Rename!!!



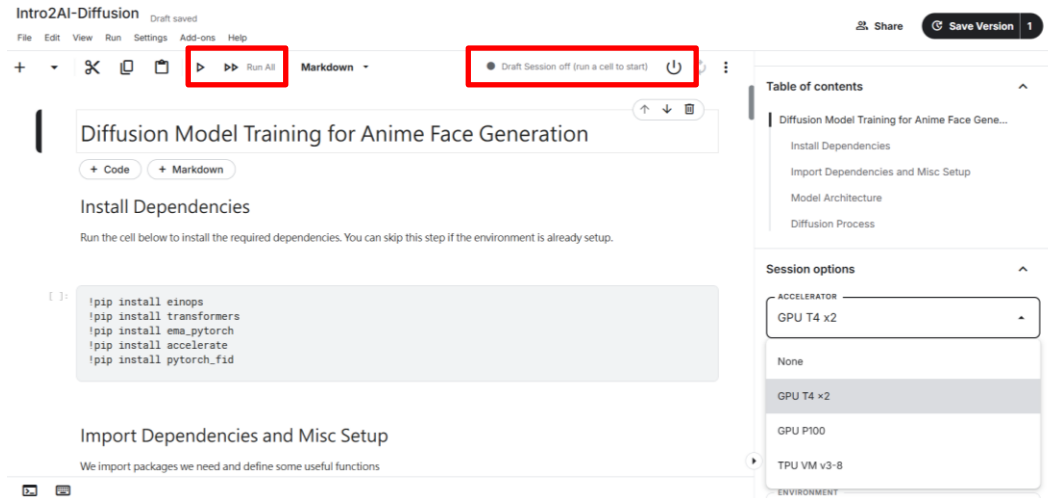
How to modify code on Kaggle

- (1) Click the code link from spec
- (2) Click “Copy & Edit” for Kaggle
- (3) Choose “GPU T4 x2”
- (4) Start session and run the code
- (5) Download Notebook and Rename!!!



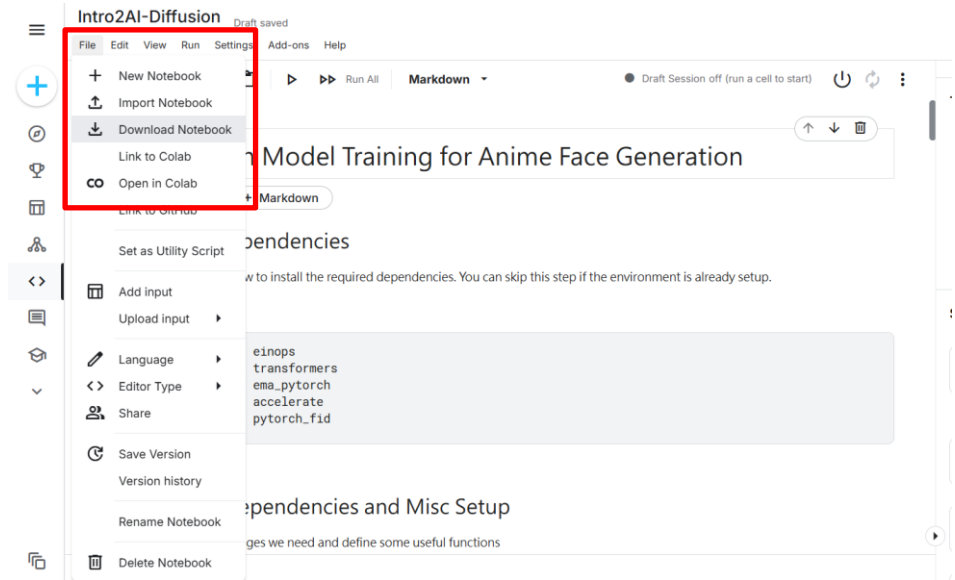
How to modify code on Kaggle

- (1) Click the code link from spec
- (2) Click “Copy & Edit” for Kaggle
- (3) Choose “GPU T4 x2”
- (4) Start session and run the code
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How to modify code on Kaggle

- (1) Click the code link from spec
- (2) Click “Copy & Edit” for Kaggle
- (3) Choose “GPU T4 x2”
- (4) Start session and run the code
- (5) Download Notebook and Rename!!!

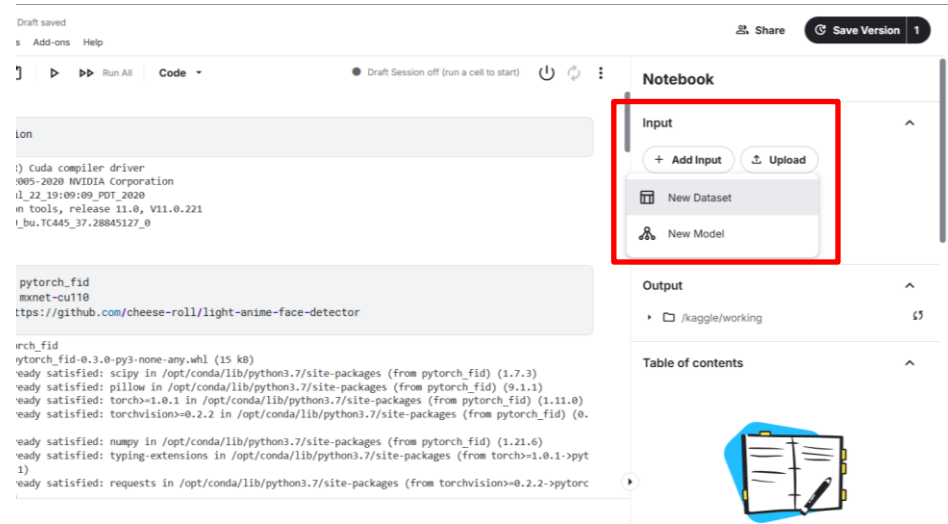


How to evaluate FID & AFD

(1) Click “Upload” and select Dataset

(2) Upload ZIP file and give a title

(3) Change the path of dataset



How to evaluate FID & AFD

(1) Click “Upload” and select Dataset

(2) Upload ZIP file and give a title

(3) Change the path of dataset

× Upload Data

DATASET TITLE

www.kaggle.com/datasets/rainsytwu/generated

VISIBILITY

FILES

Drag & drop additional files to upload
Consider zipping large directories for faster uploads

or

+ Upload another file

Reset Create

How to evaluate FID & AFD

(1) Click “Upload” and select Dataset

(2) Upload ZIP file and give a title

(3) Change the path of dataset

